Data Science Project Report — *Pediatric Appendicitis Classification*

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Abstract

Acute appendicitis remains difficult to diagnose reliably in children. Existing scoring systems, biomarkers, and imaging are each individually unreliable and negative appendectomy rates as high as 20% have been reported. We evaluate six classification algorithms: k-Nearest Neighbors, Support Vector Machines, and four tree-based methods (Decision Tree, Random Forest, XGBoost, LightGBM), for predicting appendicitis diagnosis from a publicly available dataset of 782 pediatric patients described by 53 clinical, laboratory, and ultrasound-derived features. Tree-based methods consistently outperformed distance-based classifiers, with tuned XGBoost achieving the best overall performance (accuracy 0.961, ROC-AUC 0.980), closely followed by Random Forest and LightGBM. Feature importance analysis confirmed that established clinical signs, appendix diameter and rebound tenderness, drive predictions rather than artifacts of our missing-data handling, though we identify and discuss a residual verification bias arising from which tests were ordered. An exploratory analysis of the secondary *Severity* target showed slightly weaker performance due to severe class imbalance, highlighting a direction for future work.

1 Introduction

Acute appendicitis is among the commonest childhood diseases which require surgical intervention in pediatric patients, with peak incidence between age 10 and 19 and disproportionately high perforation rates in younger patients Marcinkevics et al. [2021]. Despite being a frequent diagnosis in emergency care, appendicitis remains difficult to diagnose reliably. Its clinical presentation varies widely and classic symptoms are frequently absent, particularly in younger children Thapa Kshetri et al. [2026]. The consequences of diagnostic error are critical in both directions. Missed or delayed diagnosis increase the risk of perforation and more severe disease progression, while overdiagnosis leads to unnecessary appendectomies. Negative appendectomy rates as high as 20% have been reported Acharya et al. [2017]. Each false positive carries avoidable surgical risk and cost, while false negatives can have even worse consequences.

To support diagnosis, clinicians rely on a combination of physical examination, laboratory biomarkers, clinical scoring systems and imaging. In day to day pediatric care scoring systems such as the Alvarado Score (AS) and Pediatric Appendicitis Score (PAS) are oftentimes used Macco et al. [2016]. In addition laboratory markers such as white blood cell count (WBC) and c-reactive protein (CRP) can guide the diagnosis. Imaging modalities such as ultrasound (US) and computer tomography (CT) are used to refine the estimate further Marcinkevics et al. [2021]. Our project is also based on this work, as they kept collecting data after publishing their findings Marcinkevičs et al. [2023]. They applied machine learning to jointly predict diagnosis, management and severity in children presenting with suspected appendicitis. We focus primarily on the *Diagnosis* target, with some exploratory analysis for *Severity* and the *Management* target left for future work.

The input to our algorithms is the full dataset in its current publicly available form as a pandas dataframe with clinical measurements, laboratory markers, clinical scoring systems and imaging encoded as continuous, categorical and binary datatypes. We then explore the potential of k-nearest Neighbor Algorithm (kNN), Support Vector Machine (SVM), and several tree based methods (TBM) to predict the primary target *Diagnosis*.

2 Related Work

Diagnostic scoring systems remain a widely used decision aid for suspected pediatric appendicitis. But comparative studies show, that AS, PAS and Appendicits Inflammatory Response (AIR) score don't achieve sufficient sensitivity and specificity to be used as a standalone diagnostic standard Macco et al. [2016]. Biomarker-based approaches face a similar ceiling: WBC and CRP show only moderate discriminative power (reported area under the receiver operator characteristic curve (AUC) of 0.75 and 0.80, respectively). No single biomarker has been established as reliable decision criterion in routine practice Marcinkevics et al. [2021]. Imaging can help for diagnostic uncertainty. US is a safe and inexpensive option, but highly operator-dependent, with failure to visualize the appendix in many cases. CT offers higher accuracy, but at the cost of exposure to radiation which is particularly undesirable in children Acharya et al. [2017]. The mix of heterogeneous evidence sources is what motivates a data-driven, multivariate approach.

Most directly relevant to our work, Marcinkevics et al. [2021] introduced the dataset underlying this project. Their original cohort comprised 430 patients. Explicitly framing their study as a pilot, they already achieved good performance with a random forest classifier (RF). They achieved an AUC of 0.94 for the primary target *Diagnostic* with their dataset being substantially smaller than our updated version. This study is worth checking out, as it also laid the data foundation for more recent publications. Based on the more recent version of the datset several machine learning frameworks for pediatric appendicitis have been developed. With the bigger dataset and a clinically grounded imputer RF achieves a performance of 0.98 AUC and 93% accuracy Thapa Kshetri et al. [2026]. A even more sophisticated approach is fusing ultrasound imaging (via a ResNet-18 encoder) with structured clinical variables (via TabNet) through a cross-attention mechanism Barthwal et al. [2026]. Following this approach, their study illustrates the potential of multimodal fusion but relies on imaging data and deep architectures hard to understand in detail with limited time. While reporting a slightly lower AUC of 0.968 the multimodal approach achieves a high accuracy of 97.3%.

3 Dataset and features

The dataset comprises 782 pediatric patients presenting with suspected appendicitis. The 53 originally recorded variables describe demographics, clinical examination, laboratory biomarkers and ultrasound findings. The three targets presented with the data are *Diagnosis* (with labels no appendicitis, appendicitis), *Severity* (with labels uncomplicated, complicated) and *Management* (with labels conservative, primary surgical, secondary surgical, simultaneous appendectomy). The pandas dataframe was imported from UC Irvine Machine Learning Repository and is publicly available Marcinkevičs et al. [2023]. After preprocessing our analysis set comprises 774 patients and 82 features. The increase from 53 to 82 features comes from two sources. We added a `_observed` indicator column for variables with informative missingness (see below), and one-hot encoding expanding a nominal variable into multiple binary columns.

Not all missingness is equal. Some values are missing essentially at random and carry no information (MCAR). Others are missing not at random (MNAR). An additional test to conform appendicitis might be orderer for only very few patients where the presence of appendicitis is clinically suspected. So the fact that it was (not) ordered is itself valuable information (though it may partly reflect clinical suspcian rather than the patient's true condition, see Section 5.8). For MNAR variables simply imputing the mode for binary features or the median for continuous features would discard this signal. A binary variable that was only recorded for few patients but there most of the time shows a 1 (or a 0) could be a valuable signal for predicting the target. But filling with the mode (which is necessary for kNN and SVM to work) would bring down the predictive value of this feature close to zero. We instead add a binary `_observed` indicator (1 if recorded, 0 if missing) alongside the (filled) value, so the models can learn from missingness directly. We built two parallel preprocessed versions of the dataset, one for tree-based models and one for distance-based models (SVM, kNN), since the two families require different treatment of filled-in values (two pipelines).

Negligible missingness.
Affected patients were excluded via listwise deletion in both pipelines (782 $\rightarrow$ 777 patients).
Variables: Sex, US_Performed.
Continuous, low MNAR (essentially MCAR).
Median imputation was applied identically in both pipelines.
Variables: Age, BMI, Height, Weight, Length_of_Stay, Alvarado_Score, Pediatric_Appendicitis_Score, Body_Temperature, WBC_Count, Neutrophil_Percentage, RBC_Count, Hemoglobin, RDW, Thrombocyte_Count, CRP, Segmented_Neutrophils.
Continuous, high MNAR.
A binary `_observed` indicator was added to preserve the missingness signal. Missing values were then filled with median imputation applied after adding the `_observed` indicator, in both pipelines.
Variables: Appendix_Diameter.
Binary (yes/no), high MNAR.
A binary `_observed` indicator was added to preserve the missingness signal. Missing values were then filled with a sentinel (-1) for TBM, exploiting trees' ability to split off out-of-range values cleanly, or with the mode for kNN/SVM, since an out-of-range sentinel would distort distance-based decision boundaries.
Variables: Appendix_on_US, Migratory_Pain, Lower_Right_Abd_Pain, Contralateral_Rebound_Tenderness, Coughing_Pain, Nausea, Loss_of_Appetite, Dysuria, Psoas_Sign, Ipsilateral_Rebound_Tenderness, Appendix_Wall_Layers, Target_Sign, Appendicolith, Surrounding_Tissue_Reaction, Appendicular_Abscess, Bowel_Wall_Thickening, Conglomerate_of_Bowel_Loops, Ileus, Coprostasis, Meteorism.
Binary (yes/no), low MNAR (essentially MCAR).
No indicator was added, since missingness here is uninformative. TBM retained missing values (NaNs), allowing trees to route affected patients automatically during splitting; kNN/SVM require complete input and were mode-filled instead.
Variables: Neutrophilia, Free_Fluids, Pathological_Lymph_Nodes, Enteritis.
Ordinal, high MNAR.
Categories were mapped to integers 0–3 preserving their natural order, then handled identically to the binary high-MNAR case (indicator plus sentinel fill for TBM, mode fill for kNN/SVM).
Variables: Ketones_in_Urine, RBC_in_Urine, WBC_in_Urine, Peritonitis, Perfusion, Perforation.

Nominal.
One-hot encoded after missing-value handling (with an `_observed` indicator added first, given high MNAR), identically in both pipelines.
Variable: Stool.
Excluded.
Excluded from analysis entirely due to too few observations and different wording in free-text values, making it hard to analyze without domain knowledge.
Variables: Abscess_Location, Lymph_Nodes_Location, Gynecological_Findings.

Both pipelines preserve the MNAR signal via `_observed` indicators, but differ in how the missing value itself is filled. For binary features we used a sentinel fill for TBM. For distance-based models, we used a mode-fill for these variables since they are sensitive to out-of-range values distorting distances.

4 Methods

We evaluated six classification algorithms for predicting appendicitis diagnosis exploring simple, highly interpretable models as well as slightly more sophisticated methods which can model more complex relationships. These comprised two distance-based classifiers, k-Nearest Neighbors (kNN) and Support Vector Machines (SVM), and four tree-based methods (TBM): Decision Tree, Random Forest, XGBoost, and LightGBM. For kNN, we additionally investigated whether dimensionality reduction via Principal Component Analysis (PCA) improved performance.

For each algorithm, we first trained a baseline model with default (or otherwise standard) hyperparameters. Then we trained a tuned version selected via GridSearchCV (kNN, SVM, Decision Tree) or RandomizedSearchCV (Random Forest, XGBoost, LightGBM, given their larger hyperparameter spaces), using five-fold cross-validation on the training data throughout to ensure a fair comparison while preventing overfitting and data leakage. A detailed description of each algorithm is provided in Appendix A.1. For specific hyperparameters for the tuned models see A.3.

5 Results

To assess the classification performance of the evaluated models, we considered several evaluation metrics, including accuracy, precision, recall (sensitivity), F1-score, and the area under the receiver operating characteristic curve (ROC-AUC). Since the task involves medical diagnosis, recall is of particular importance, as false negative predictions correspond to undetected cases of appendicitis and may lead to delayed treatment. The following sections compare the baseline and optimized versions of the evaluated models and discuss the effect of hyperparameter tuning and dimensionality reduction on their predictive performance.

5.1 k-Nearest Neighbors (kNN)

Hyperparameter optimization evaluated different values of k, Euclidean and Manhattan distance metrics, and uniform and distance weighting using GridSearchCV. The best-performing model used 11 neighbors, Euclidean distance and uniform weighting, improving the accuracy from 78% to 83% (Table 1). The larger neighborhood likely reduced sensitivity to noisy observations while preserving local decision boundaries. Permutation feature importance identified glucose and BMI as the most influential predictors, whereas age contributed least to the final predictions.

5.2 Principal Component Analysis (PCA)

To evaluate whether dimensionality reduction benefits distance-based classification, PCA retaining approximately 95% of the total variance was applied prior to kNN classification. The optimized PCA-kNN model achieved 80% accuracy, below the original kNN classifier (Table 1). These results suggest that preserving overall variance does not necessarily improve class discrimination, as informative low-variance features may still contribute to predictive performance.

5.3 Support Vector Machines (SVM)

GridSearchCV evaluated both linear and RBF kernels together with the regularization parameter C. The optimized model selected an RBF kernel, improving the accuracy from 81% to 84% while achieving the highest ROC-AUC (0.91) among the distance-based models (Table 1). This suggests that nonlinear decision boundaries better captured the underlying data structure. Permutation feature importance again highlighted glucose and BMI as the dominant predictors, with age contributing comparatively little.

5.4 Decision Tree

The tuned decision tree achieved an overall accuracy of 0.916, already matching the SVM classifier despite being the simplest and most interpretable model among the algorithms explored. It obtained a precision of 0.916 and a recall of 0.946 for patients diagnosed with appendicitis, corresponding to 5 false negative predictions, with a ROC-AUC of 0.90. Despite its simplicity, the tuned decision tree already performed similar to the distance-based classifiers above, indicating the stronger performance of the ensemble methods that follow.

5.5 Random Forest

The tuned RF substantially outperformed the decision tree on the held-out test set, achieving an overall accuracy of 0.955, a precision of 0.957, and a recall of 0.967 for the appendicitis class, corresponding to only 3 false negative predictions and a ROC-AUC of 0.979. This improvement is consistent with the expectation that aggregating many decorrelated trees reduces the variance and overfitting tendency of a single decision tree. This performance together with a manageable model complexity is one of the reasons why it is used by others for diagnostic classification Marcinkevics et al. [2021], Thapa Kshetri et al. [2026].

5.6 XGBoost

The tuned XGBoost model achieved an overall accuracy of 0.961, a precision of 0.957, and a recall of 0.978 for the appendicitis class, corresponding to only 2 false negative predictions and a ROC-AUC of 0.980. Across our entire set of six classifiers, the tuned XGBoost model achieved the best performance on every evaluation metric on the held-out test set (see Table 1). Its sequential, error-correcting boosting procedure and build-in regularization allow it to capture complex, non-linear relationships between clinical features, while still controlling for overfitting.

5.7 LightGBM

The tuned LightGBM model achieved a slightly lower accuracy on the test data of 0.948, a precision of 0.957, and a recall of 0.957 for the appendicitis class, corresponding to 4 false negative predictions and a ROC-AUC of 0.966. Underperforming XGBoost across all metrics, LightGBM's leaf-wise growth strategy, while efficient, may have made it slightly more prone to overfitting on this particular training split than XGBoost's regularized, depth-constrained boosting procedure. Since all test-set scores fell within the range expected from the cross-validation standard deviation, this gap can easily be interpreted as ordinary variance rather than methodological error.

5.8 Feature Importance and Limitations

We found that for the TBM classifiers, *Appendix_Diameter* and *Ipsilateral_Rebound_Tenderness* were consistently important across all models. Both are genuine clinical measurements, not `_observed` indicators, suggesting our models rely on true clinical signal. Removing all `_observed` indicators and only working with the unaltered original dataset Marcinkevičs et al. [2023] (TBM models handle missing data natively) caused only a very small drop in test accuracy, from 0.961 to 0.955, indicating this signal is not highly relevant as we even get an increase in recall and ROC-AUC for the best performing model which was tuned XGBoost for this unaltered dataset as well (recall of 0.989, ROC-AUC of 0.987). As noted in Section 3, `_observed` indicators partly encode the referring clinician's prior suspicion rather than the patient's physiology alone, a pattern known as *verification bias*. This is not data leakage as a real clinician would know the same information. But it means our results reflect a post-workup setting rather than early triage. The exploratory analysis for the Severity target performed slightly worse than the Diagnosis target, also driven by strong class imbalance. For details see A.2.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Baseline kNN	0.781	0.830	0.793	0.811	0.867
Tuned kNN	0.826	0.874	0.826	0.849	0.906
PCA + Tuned kNN	0.800	0.802	0.880	0.839	0.907
Baseline SVM	0.916	0.934	0.924	0.929	0.961
Tuned SVM	0.897	0.913	0.913	0.913	0.961
Decision Tree (baseline)	0.910	0.906	0.946	0.926	0.901
Decision Tree (tuned)	0.916	0.916	0.946	0.930	0.900
Random Forest (baseline)	0.955	0.957	0.967	0.962	0.969
Random Forest (tuned)	0.955	0.957	0.967	0.962	0.979
XGBoost (baseline)	0.955	0.947	0.978	0.963	0.977
XGBoost (tuned)	0.961	0.957	0.978	0.968	0.980
LightGBM (baseline)	0.955	0.957	0.967	0.962	0.968
LightGBM (tuned)	0.948	0.957	0.957	0.957	0.966

Table 1: Comparison of all explored models predicting the primary target *Diagnosis*

6 Conclusion

We evaluated six classification algorithms for predicting pediatric appendicitis diagnosis, finding that TBMs outperformed the distance-based classifiers. XGBoost achieved the best overall performance (accuracy 0.961, precision 0.957, recall 0.978, ROC-AUC 0.980), closely followed by Random Forest and LightGBM, while even a single decision tree got similar results to the SVM. This performance is not far off from the recently reported 97.3% which used a substantially more complex multimodal fusion approach combining ultrasound imaging and clinical data Barthwal et al. [2026]. While further research with complex models which evaluate the imaging data directly might have the potential to achieve even better performance (with more data), our approach suggests that structured clinical features alone already capture much of the diagnostically relevant signal. This helps with interpretability and explainability of the ML pipeline and might be a good intermediate solution for settings where more complex models can't be applied due to missing images or other data concerns. Feature importance analysis showed that our models rely substantially on established clinical signs rather than solely on missingness artifacts. We still identified a residual verification bias tied to which diagnostic tests were ordered, meaning our results are best understood as reflecting a post-workup diagnostic setting. Future work should extend this pipeline to the *Severity* and *Management* targets, both of which are not sufficiently explored here.

Contributions

- **Mathis Hasler:** Did the research regarding the related work and the dataset. Spend a lot of time on preprocessing the data for further analysis with all models. Implementation and evaluation of all Tree-based methods used. Wrote all sections of the final report except the parts that covered kNN and SVM models.
- **Korbinian Staudacher:** Worked on Assignment 2, training and evaluating models for regression and classification tasks.
- **Carin Sabah:** focused on implementing and evaluating the kNN and SVM models, including hyperparameter tuning and a PCA-based extension for kNN. Also analyzed the experimental results and documented the methods and findings in the final report.

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A Appendix

A.1 Detailed description of the methods explored

The first algorithm considered was k-Nearest Neighbors (kNN), a simple distance-based, non-parametric classification algorithm that assumes that observations which are close in the feature space are also likely to belong to the same class. As a baseline implementation, we trained a kNN classifier using five nearest neighbors (k = 5), Euclidean distance as the distance metric, and uniform voting, where each neighbor contributes equally to the final prediction.

To investigate the influence of different hyperparameters, we additionally trained a fine-tuned version of the model using GridSearchCV. The search considered different values of k, both Euclidean and Manhattan distance metrics, as well as uniform and distance-weighted voting schemes. Since kNN directly relies on distances between observations, we further investigated whether Principal Component Analysis (PCA) could improve predictive performance by reducing redundancy and noise in the feature space before classification.

Our second algorithm was the Support Vector Machine (SVM), which learns a maximum-margin separating hyperplane between the classes. In our case, the prediction target was the diagnosis (appendicitis vs. no appendicitis), meaning that the objective of the classifier was to separate patients according to their diagnosis.

Similar to kNN, we additionally trained a tuned SVM model using GridSearchCV. Hyperparameter tuning explored different values of the regularization parameter C, which controls the trade-off between maximizing the margin and penalizing classification errors. Furthermore, we evaluated both linear and radial basis function (RBF) kernels to investigate whether allowing nonlinear decision boundaries improved predictive performance.

Turning to tree-based methods, we first considered the Decision Tree. It is a simple and highly interpretable algorithm that recursively splits the feature space along individual variables to produce a set of hierarchical decision rules. As a baseline, we trained an unconstrained decision tree, allowing it to grow without depth restrictions. To control overfitting, we additionally trained a tuned version using GridSearchCV, opimizing over tree depth, the minimum number of samples required to split a node, and the minimum number of saumples required at a leaf.

Our second tree-based method, which was also used by Marcinkevics et al. [2021] was the Random Forest (RF). It's an ensemble method that aggregates predictions from a large number of decision trees, each trained on a bootstrapped sample of the training data with a random subset of features considered at each split. By averaging over many decorrelated trees, RFs reduce the variance and overfitting tendency of a single decision tree. Although they can't be visualized easily like a single Decision Tree, feature importance can still be extracted at the ensemble level. We tuned the RF using RandomizedSearchCV due to the larger hyperparameter space, exploring the number of trees, maximum tree depth, minimum samples per split and leaf, and the number of features considered at each split.

We further considered two gradient boosting algorithms, XGBoost and LightGBM. Both build an ensemble of trees sequentially, with each new tree trained to correct the residual errors of the previous ones. This sequential correction typically yields higher predictive performance than bagged ensembles, at a further cost to interpretability. XGBoost incorporates regularization terms to penalize model complexity. LightGBM employs a leaf-wise tree growth strategy together with histogram-based split-finding, which improves training efficiency on high-dimensional data. For both algorithms, we trained a baseline model with default hyperparameters and a tuned version via RandomizedSearchCV. The fine-tuning explored the hyperparameters number of estimators, learning rate, maximum tree depth and subsampling ratios for both observations and features.

A.2 Severity Assessment

The class proportions for the target *Severity* in the dataset were 0.851 for label 0 (uncomplicated) and 0.149 for label 1 (complicated).

Model	**Accuracy**	**Precision**	**Recall**	**F1-score**	**ROC-AUC**
Decision Tree (baseline)	0.916	0.727	0.696	0.711	0.825
Decision Tree (tuned)	0.935	0.760	0.826	0.792	0.941
Random Forest (baseline)	0.923	0.789	0.652	0.714	0.967
Random Forest (tuned)	0.948	0.826	0.826	0.826	0.968
XGBoost (baseline)	0.942	0.792	0.826	0.809	0.971
XGBoost (tuned)	0.942	0.769	0.870	0.816	0.969
LightGBM (baseline)	0.948	0.826	0.826	0.826	0.971
LightGBM (tuned)	0.942	0.792	0.826	0.809	0.972

Table 2: Comparison of TBMs predicting the secondary target *Severity*

A.3 Hyperparameters for the different models

Model	Hyperparameter	Value
kNN	n_neighbors	11
	weights	distance
	metric	euclidean
PCA + kNN	n_components	5
	explained variance	0.314
	n_neighbors	15
	weights	distance

Table 3: Hyperparameters used for the kNN models.

Model	Hyperparameter	Value
SVM	C	0.1
	kernel	RBF
	gamma	scale

Table 4: Hyperparameters used for the SVM model.

Model	Hyperparameter	Value
Decision Tree	max_depth	10
	min_samples_split	2
	min_samples_leaf	1
	ccp_alpha	0.002

Table 5: Hyperparameters used for single decision tree.

Model	Hyperparameter	Value
Random Forest	n_estimators	355
	max_depth	15
	min_samples_split	2
	min_samples_leaf	1
	max_features	0.5
	bootstrap	True

Table 6: Hyperparameters used for Random Forest.

Model	Hyperparameter	Value
XGBoost	n_estimators	352
	learning_rate	0.185
	max_depth	7
	subsample	0.832
	colsample_bytree	0.640
	min_child_weight	3
	gamma	0.206
	reg_alpha	0.941
	reg_lambda	1.333

Table 7: Hyperparameters used for XGBoost.

Model	Hyperparameter	Value
LightGBM	n_estimators	322
	learning_rate	0.153
	num_leaves	14
	max_depth	9
	min_child_samples	18
	subsample	0.882
	colsample_bytree	0.546
	reg_alpha	0.737
	reg_lambda	1.092

Table 8: Hyperparameters used for LightGBM.